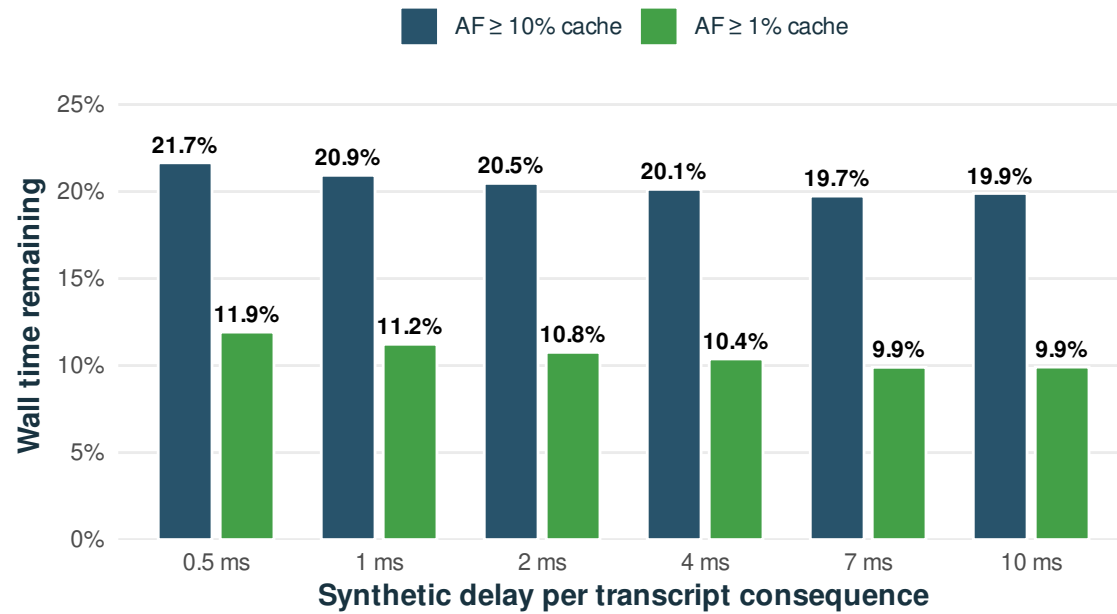


Bundled cache coverage determines how much pipeline time returns

One 4.80-million-variant real WGS · matched direct baselines · bundled Zenodo caches · light statistics · no technical repeats

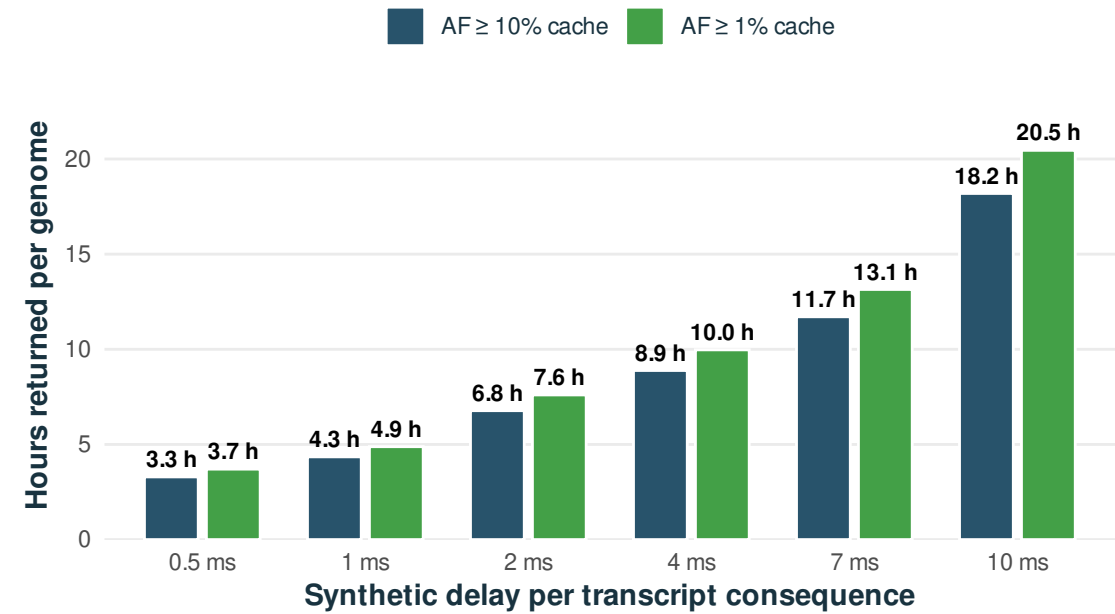
A More cache coverage removes more waiting

Measured share of direct wall time remaining



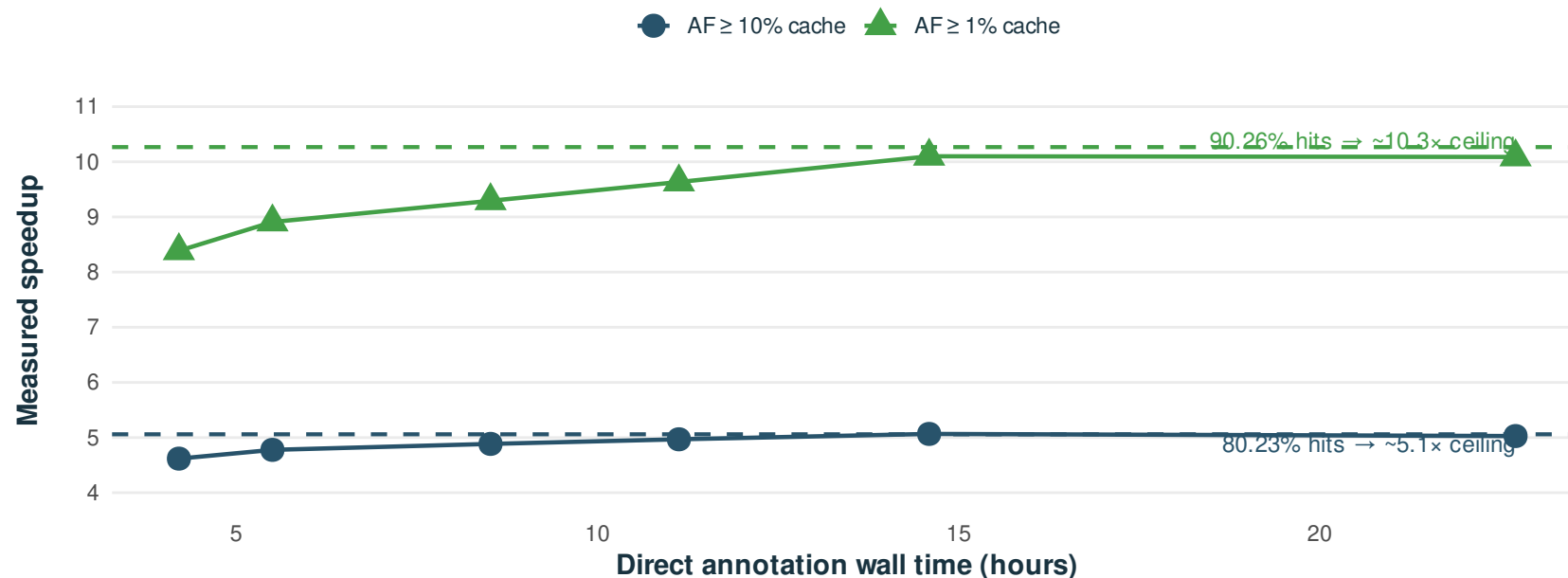
B Absolute savings grow with pipeline cost

Up to 20.5 hours returned for one real WGS



C Speedup approaches the hit-rate ceiling

The larger bundled cache nearly doubles the slow-pipeline ceiling



D One runtime model explain

Lookup overhead + miss fraction × dire

